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EX9 Implementation of Subnetting in CISCO packet Tracer Simulator

AIM:

To implement subnetting in CISCO packet tracer simulator.

PROCEDURE:

1. Creating a network Topology

The first step in implementing classless IP subnetting is to create a network topology in Packet Tracer. To create a network topology in Packet Tracer, select the "New" button in the top left corner, then select "Network" and "Generic". This will create a blank network topology that we can use to add devices.

2. Adding the Devices

Once we have created our network topology, we can add devices to it. Here, we will be adding routers, switches and PCs. To add a device, select the device, select the device from the bottom left corner and drag it onto the network topology. Then, connect devices by drawing a cable from one device's port to another device's port.

3. Subnetting.

To subnet the network address of 192.168.1.0/24 to provide enough space for atleast 5 addresses for end devices, the switch and the router, we can use a /27 subnet mask. This will give us 8 subnets with 30 host addresses each.

The IP addressing for network shown in topology can be as follows:

- Router R1:
- Gigabit Ethernet 0/0: 192.168.1.1
- Gigabit Ethernet 0/1: 192.168.2.1
- Switch S1:
- Fast Ethernet 0/1: 192.168.1.0/27
- PC1: 192.168.1.11
- PC2: 192.168.1.12
- PC3: 192.168.1.13
- PC4: 192.168.1.14
- PC5: 192.168.1.15
- Fast Ethernet 0/2: 192.168.2.0/27
- PC1: 192.168.2.11
- PC2: 192.168.2.12
- PC3: 192.168.2.13
- PC4: 192.168.2.14
- PC5: 192.168.2.15

Router 2:

- Fast Ethernet 0/1: 192.168.3.0/27
- PC1: 192.168.3.11
- PC2: 192.168.3.12
- PC3: 192.168.3.13
- PC4: 192.168.3.14
- PC5: 192.168.3.15
- Fast Ethernet 0/2: 192.168.4.0/27

PC1: 192.168.4.11
PC2: 192.168.4.12
PC3: 192.168.4.13
PC4: 192.168.4.14
PC5: 192.168.4.15

4. Configuring the Devices:

Now that we have added our devices and connected them, we can start configuring them. We will start by configuring router. Right click on router and select "CLI". This will open command line interface for the router. In the CLI, enter following commands.

```
# enable
# configure terminal
# interface Fast Ethernet 0/0
# ip address {IP address} {subnet mask}
# no shutdown
# exit
interface Fast Ethernet 0/1
ip address {IP address} {subnet mask}
no shutdown
exit
```

Replace "{IP address}" and "{subnet mask}" with your desired IP address and subnet mask. The first interface, Fast Ethernet 0/0, will be connected to the switch, while the second interface, Fast Ethernet 0/1 will be connected to one of the PCs. These commands configure the router's interfaces with IP addresses and subnet masks.

Next we will configure the switch. Right click on switch and select "CLI". In the CLI, enter the following commands:

```
enable
configure terminal
interface FastEthernet 0/1
switchport mode access
exit
```

interface 'Fast Ethernet 0/2
switchport mode access
exit

These commands configure the switch to operate in access mode on its two ports, which are connected to two PCs.
Finally we will configure the PCs. Right click on each PC and select "config". In configuration window, enter IP address, subnet mask, default gateway, and DNS server information. The IP address and subnet mask should be within same subnet as router's Fast Ethernet 0/1 interface

To configure GigabitEthernet interface on the router, you can follow these steps:

1. Right click on the router and select "CLI".
2. Enter the following commands:

```
enable
configure terminal
interface GigabitEthernet 0/0
ip address {IP address} {subnet mask}
no shutdown
exit
```

Testing the network:

Now that our network topology is configured, we can test the network. Open a command prompt on each PC and try to ping the other PC. If the ping is successful, then network is functioning properly. We can use "ping" command to test connectivity between router and the PCs.

RESULT:

Thus the subnetting in CISCO packet Tracer simulator is implemented.